

ActInf Livestream #020.0 ~ The Emperor's New Markov Blankets

Livestream | The Emperor's New Markov Blankets | 1:31:30

hello and welcome
to the active inference lab this is
active inference live stream number 20.0
on april 15
2021 i'm daniel and i am here with blue
who will say hello
hi i'm blue knight i'm an independent
research consultant out of new mexico
thanks for being on blue and looking
forward to this conversation
welcome to the active inference lab
everyone we are a participatory online
lab
that is communicating learning and
practicing applied active inference
you can find us at the links here on
this page
this is a recorded and an archived live
stream so please provide us with
feedback
so that we can improve on our work all
backgrounds
and perspectives are welcome here and
we'll be following
good video etiquette for live streams
20.0 is a contextualizing stream
for the upcoming discussions that we'll
have on april 20th
and 27th about this paper
the emperor's new markov blankets and
we'll have some of the authors join us
as well so this will be an awesome
discussion here we are in
number 20.0 and the goal of this
discussion as stated is to set the

context for the upcoming
discussions that we'll have on number 20
and number 20.1
and that is about the paper the
emperor's new markov blanket
by yella brunberg at all and the link is
provided here
the video is just an introduction and us
kind of working through
the paper so that we're ready to
interact with authors and
really grasp what they're discussing and
we'll be going through the usual
sections of the paper um
any uh just overall thoughts before we
sort of lead into it blue just what did
you
like about it or what drew you to the
paper
i liked this paper i thought it really
provided an overarching view
of the markov blanket and its
use throughout like historical use and
also throughout the the first in context
so i thought it was really thorough and
gave a good treatment to
all of those topics nice agreed
so the paper is emperor's new markov
blankets
and the links provided here the aims of
the claims of the paper that's where
we're going to start with the aims and
the abstract and the roadmap
to just see how the authors laid out
what they're going to be
trying to do so we're going to evaluate
their
aims and then see whether the claims

they make are valid
and whether they support their aims that
they laid out
and they write i'll read the first one
blue
the aim of the paper is twofold first we
want to explain
how it has been possible for such an
innocuous
technical concept as a markov blanket to
come to be used in order to settle
central debates
in the philosophy of biology and
cognition
so that's the main key aim here is how
did we get here
how did we get to debating a statistical
technical concept like the markov
blanket and having people talk about
consciousness and emotion
and multi-skill systems so how did we
get here with the markov blanket
and then blue you can go for the second
name and what you thought it meant
they say we will trace the development
of markov blankets
starting from their standard application
in bayesian networks
through the role that they play in
variational inference to use
to their use in the literature on active
inference which is what i had said about
the paper that i liked
they say we will argue that in the
course of this transition
a new and largely independent
theoretical construct has emerged
one that is more closely aligned with

the notions of sensory motor loops
and agent environment boundaries do you
have any thoughts on that daniel
it's gonna be the core distinction that
we're going to hit probably 50 times
is that there's two ways to talk about
boundaries maybe the ones that really
are there and then the ones that we
infer
and that's what this paper's about
so here we go to the abstract
markov blankets have been used to settle
disputes central to
philosophy of mind and cognition their
development from a technical concept
in bayesian inference to a central
concept within the free energy principle
is analyzed
so that's the historical component of
the paper
then there's the sort of distinguishing
part of the paper
we propose to distinguish between
instrumental pearl blankets and realist
fristen blankets pearl blankets are
substantiated by the empirical
literature but
can do limited philosophical work first
in blankets can do philosophical work
but require strong theoretical
assumptions
both are conflated in the current
literature on the free energy principle
and then their conclusion is
consequently we propose that
distinguishing
between an instrumental and a realist
research program will help clarify

the literature
good good on to the road map which now
has a
55 miles per hour speedway on the left
side
for the rapid run through they
start with introducing bayesian
inference
and really walk through it in a nice way
that we're going to walk through
as well and go from kind of the simplest
form of bayesian inference to
surprisingly
nuanced forms that introduce
networks like bayesian networks and
that's where we get to this concept of a
pearl blanket
and then it's by tracing the history and
the citations and the formalisms around
the pearl blanket and the markov blanket
that we find out where it intersects
with active inference
which is where the authors propose that
this fristen blanket
concept sort of surreptitiously or
without being noticed
starts getting deployed in a way that's
um sort of sometimes one way sometimes
the other way
but it's actually the looseness around
the definition that in their opinion
gives rise to
uncertainty which is not what we're
about here
and in section
five they really just provide more
examples with
walkthroughs and graphs and figures that

we're going to walk through today
with how is the pearl blanket different
than the first and blanket
sense of the markov blanket and then how
does that relate to this
key distinction at the end between
inference with a model
versus inference within a model so kind
of like generative models versus
generative processes
that's going to come back and um
it's a great paper so we hope that you
listen to these
discussions and also read it any
thoughts on the road map nope it's going
to go fast
yes it's a long period yeah so
the title of the paper uh is an allusion
to the story called the emperor's new
clothes
so people can read through the way that
it's written
here but um the emperor's new clothes
is a interesting story it's a polycymous
story there are many meanings of it so
what is the emperor's new clothes about
blue in general or what
what is it doing here in the title and
we can definitely ask the authors but
what do you think it's doing here
so in my understanding of the emperor's
new clothes it's definitely like
without reading the screen it's a story
about an emperor who wants who goes to
like the best
you know clothes maker in town gives
them all kinds of money
and the clothes maker proceeds to make

them clothes that are invisible and then
when the emperor he
sees essentially maybe what he wants to
see in the clothes like sees this you
know and dresses himself in them and
then goes parading throughout the town
naked

because the clothes are invisible
and then the child says hey that guy's
naked

yeah and then here's what part i had
forgotten was although startled
the emperor continues the procession
walking more proudly than ever
so it's like it's not like oh and then
he was so embarrassed that he stopped
doing it so

maybe it's a story we'll keep in mind
and we'll ask who are these
characters or what do the authors mean
by that

but also at this sort of general level
why

why does this paper matter why are we
talking about it and studying it and
discussing the implications
because it's good to be clear when we're
talking about

science or philosophy and

uh an unfortunate reality of science is
that

it's easy to slip in metaphysical claims
into

uh so-called facts of the world or even
just measurements of the world

and it's not good to slip in ungrounded
metaphysical claims

and then also this realism

instrumentalism distinction
is something that has been dealt with
basically
in every field oh is it the model of
language you're talking about or are you
really talking about language or
those kinds of differences between
systems as they are and systems as we
model them
is something that is a nomadic concept
so to see
it appear in active inference is cool
but also it's exciting to be involved
with the authors and see
how the scientists and the philosophers
are actually
dealing with this question in a new way
at a new time scale
because of maybe the way that they're
communicating with each other
the big questions in the paper like sort
of beyond
just what they set out to address would
be like how is physics related to
metaphysics
what what can we study about electrons
and temperature and
mass that would tell us anything about
whether a spirit exists or
what's really out there beyond plato's
cave how is scientific modeling related
to reality
and then a big question for a lot of us
and a lot of people outside of our
active inference community
is like what does the free energy
principle and active inference
really say about pragmatic concerns

and then totally independently what do they say about philosophical concerns and it'd be awesome if they were 10 out of 10 on both but

what is the actual reality of what these frameworks tell us about pragmatic concerns and philosophical questions

what big questions resonated with you blue

uh the biggest question really that resonated with me i think is is are we you know using the right tools to study the right

things so i think that you know when we think about um

active inference and cognition and statistics like are these all

you know should these all be represented by a markov blanket like is this the right

the right thing to study this with i don't know

how how unified do we want models and systems to be

on that this is sort of the last little general

signpost before we plunge into the free energy principle and

active inference and everything it's just a little distinction

on maps and territories because realism and instrumentalism

it sounds really philosophical and there's isms and it sounds like a belief system

but it really boils down to this map territory distinction

and so the authors wrote in our view the
free energy principle
literature so whatever it is even if
you've never heard of this idea
it consistently fails to distinguish
between the map
a representation of reality and the
territory reality itself
this slippage becomes most apparent in
their treatment of the concept of a
markov blanket
so if we're ever getting lost in the
weeds with Fristen and Pearl and
instrumentalism realism
we think like map in territory and then
what's really cool about
this topic especially in the active
inference
setting is that two of the papers that
are cited in this paper
as relevant prior work are Mel
Andrews's paper and also Thomas van
den Hilbarto's paper
and these were on active inference 14
and 15 and now we're here in number 20
so it's kind of like
we're going quickly and we're speaking
with a lot of the authors
and then also on the left is um a paper
that I co-authored with
RJ and another colleague Mikhail and we
kind of stepped back
even further than active inference and
just asked
what does it mean to map across systems
and we highlighted that sometimes when
mapping across systems and using
the complexity science approach people

use bayesian statistics
networks and predictive and counter
factual approaches
those are kind of like three key ideas
in making some
systems similar to each other or
comparing different systems
and those are um citations to the active
inference
literature because active inference is
drawing together
a lot of these different domains that
might be seen as different like bayesian
statistics
network science cybernetics ecological
psychology
i mean there's different words because
they're different areas but we're
actually coming at it in a new way
and that is why we're thinking about
maps and territories in a new way
because it's not just like cognitive
maps yes or no
or maps in the city yes or no it's a lot
of new
things coming together which is why it's
good to clarify
adding on to that uh is the fields and
glaze book paper that we did
that um you know mapped the the two
spaces right from one to the other
like the cat the category theory yes
nice the mapping
and so and then what if the map is the
territory
so let's get to fep
so for this breakdown every time in
reading a paper we get to

reintroduce the free energy principle
because it's always good to start
with sort of a first pass
on what it is because this video might
be someone's
first time hearing about free energy
principle and also it's a good
opportunity to see how the authors
frame free energy principle and then use
that as a scaffolding for our
understanding and our curiosity instead
of us trying to get to the best
explanation week after week after week
we can
develop our understanding week by week
but then always be inspired by the way
that authors have phrased it in their
writing
so on the right side with the colors
i kind of broke down into like almost
haiku
or just short sentence a little bit what
they were
saying and what it comes down to is that
the fep
is a normative mathematical framework so
normative like it says what should
happen like evolution by fitness
maximization
says that it should be maximizing its
fitness that's kind of similar
it's not saying what you should do but
it's saying how systems should be
and it's about systems that are
self-organizing
and how they fit into their niche
fep is drawing on physics inspired
machine learning

approaches and so part of the appeal and the excitement of the free energy principle is that it brings together machine learning and physics and math with also potentially some qualitative dimensions but even without dipping into the qualitative or the aesthetic or the embodied or the inactive at the very least from a control theoretic perspective the free energy principle adds in a lot of features that are absent from other frameworks like network theory alone bayesian alone or predictive systems alone combining those types of approaches in a really interchangeable way with system properties like learning attention and action planning is very powerful active inference is a theory that is a corollary to or it's a compatible implementation of systems that are broadly construed under the free energy principle and then they kind of close the authors do with the recognition of a of a claim that you'll see in a lot of places that if we could have a theory of everything or a multi-scale system then it would be great so that's what motivates people to work on the fep and why a lot of people are excited about it

anything to add on that blue okay
it would be great i'm excited about it a
unifying theory of everything let's get
to it
let's get to it now we wanna just like
you said with are we studying the right
systems with the right tools
so this is what the fep is and um very
um intentionally it appeals to things
like control theory and
measurement let's take it into the deep
end with metaphysics
so the author's right the fep has been
applied to
metaphysical questions so these are very
different than
action planning learning attention these
are going to be
related to questions that are
metaphysical like what are the
boundaries of the mind and the citations
are in the paper
what are the boundaries of living
systems so what's really there
ontologically what's the relationship
between mind and matter
like what is experience all these kinds
of philosophical
questions about qualia um and even
things like is
gaia real or is there a scientific uh
basis for talking about gaia or sort of
earth level
awareness or at what level could
awareness exist
and more we're probably all curious
about a lot of questions and
it's not like we need to a priori

classify them into physics or metaphysics
um but that shouldn't make us think there's no distinction
the questions that come to our mind we're curious probably about a lot of stuff
so we want to be open to questions as they come in
especially ones that are framed in new ways and then
also be trying to approach them rigorously
because as they raise here it's a web of formalisms it's true
that it is a web because we're gonna walk through it in the order that the authors laid it out
but it is a web of concepts and so even for those who are experienced in the field but especially those who are new to the area or as they call it a layperson
not only are there distinctions that are unconventional but their implications are not always obvious like assumptions or axioms or relationships that have maybe very non-obvious conclusions or implications
especially when combined and then the two risks here are it might appear to be and actually mathematically unjustified
but at the very least appear to be and that might warrant the unintended smuggling in of metaphysical assumptions

like i have information theory i can
calculate it and here's who's conscious
and who's not conscious
i would say that's completely
unwarranted
what do you think good example yeah i
think that's i mean that's like the main
critique right of of iit
so i don't know i think that it's a good
example and
how can we say who's conscious and who's
not when you can't even give
consciousness an
adequate definition i think i don't know
but we're solving unanswerable questions
but that's the joy of it
cool yep we talked about the
consciousness side in
some of our other discussions but
suffice to say that there's kind of a
distinction between the questions on
this slide which are related to
stochastic systems and action in the
niche
and then the metaphysics which are
things that you could imagine
even robots would debate over like
who is really a robot like us or
whatever they ask
so how does the fep relate to active
inference
they write active inference is a process
theory
derived from the application of
variational inference to the study of
biological and cognitive systems
and we're not going to read this whole
piece but

active inference is a process theory so
it specifies how processes play out
which we're going to go into a lot of
detail
and it sort of takes something that's
initially static
and it figures out how to take a bunch
of equations just there on the page
and actually link them in a way where it
helps us think about how systems
work in their flow so we take a bunch of
equations that relate variables
like estimates and action states and
sensory states
that are all happening at once as
analytical equations
in the fvp and then active inference is
a
compatible architecture like a skeleton
or a style of model that implements
processes compatible with fep
what would you say about that
i think that you summed it up well uh i
mean the authors you have a couple
points here that are highlighted so
the fundamental imperative of active
inference is to minimize free energy
so this is the fep the free energy
principle
um and and then the other point is that
uh you know
this is also the the question that i had
at the beginning are we using the right
tools to study these
these phenomena right more specifically
we can ask what role
the pearl blankets might play in active
inference

so yep we're going to be seeing how this technical concept entered the discussion of metaphysical questions and it's almost like active inference as applied to biological creatures

is where we see a lot of it coming into play

and yep we talk about active inference throughout so we'll kind of return to it so now let's that's like the modern stuff right

fep and active inference let's pull back to base

and to bayesian statistics and build up in a new way how we're going to go from simple base

to modern base so

blue what would you how would you start us off or what do we need to know as we go on this like little base

uh several slide journey so like the one sentence or maybe like this the short summary that i give about

baze like bayesian statistics versus frequent

statistics and frequent to statistics which is like what

most people are taught um in school like when you flip a coin the probability of it coming up heads

half right 50 percent of the time it comes up half

bayesian statistics accounts for the fact that like the previous

12 flips have all been heads right

like so what would you if you have flipped a coin and you know it's a fair coin if it's come up heads 12 times in a

row like

the next time you flip it don't you

think it's more likely to come up tails

right like so the bayesian accounts for

what has happened

in the past uh when it's it's not

accounted for in frequent

statistics at all nice and

um just to go through the notation

because we are going to be kind of going

through the notation as they write it's

a recipe

for calculating the posterior

probability

so that's the probability of x given y

and

x is the hidden state of the world

like let's just say that the hidden

state of the world is the true

temperature

um it uh it's actually a hypothesis

about the world

that's why bay's in bayesian statistics

it's like a hypothesis that you're

testing

and uh the observational data the

measurement is

y and this is uh

posterior two data so like prior and

posterior are referring to when

new information comes in prior to the

new data coming in

and then it's updated and then you have

a posterior which has been updated

by the new data and

in very rapid here

in this paragraph they make the move

from doing

inference basing inference on a hidden variable like the temperature of the world

into this idea of the model evidence with a normalization that keeps things adding up to one

so instead of having like an uncertainty distribution of how the temperature might be in the world and then taking in new data and updating that

another way might be um i have two hypotheses it's colder than 20 or it's warmer than 20. and then i can always be judging relatively which one of those is more accurate so even if the temperature distribution is like very gnarly

it's possible sometimes to have a simpler thing that you're trying to hypothesize about the world like is it over or under a certain number that helps you guide action and then also when you do this model um comparison you can keep the relative uh basically the the relativation you can keep models being compared relative to one another

and that can sum to one so you can do probability type estimates on different model classes even without knowing whether any specific model is likely or unlikely so like in phylogenetics which is an area that blue and i have both worked you can use bayesian statistics

to
relatively evaluate what the most likely
tree is for species but that tree's
likelihood might be like
super low but that's not what matters
it's
whether it's more likely than other
hypotheses
so that's kind of what they write a
little bit about here
is when you have types of models
that are computationally or analytically
intractable so there's no closed form
solution for the posterior
like there's no clean way how you're
going to update
your model of the phylogenetic tree
given the next new genome that's added
in
it's just not something smooth like
signal processing might be
so how are you going to deal with that
and that's the big challenge
so it's kind of this rapid bayesian
acceleration from like
you're trying to estimate something
about the world that's unknown okay well
what's unknown
is which model is better and then the
challenge is
we want to update our ongoing estimate
of what the best model of the world is
but there aren't really update rules
for how new data coming in should help
us update which model it is
like night and day it's clear more light
it's likely to be daytime
but again think about the phylogenetics

or something where as new data comes in
it's really hard to know how exactly to
update your models
so in that uh the phylogenetic analysis
if you have new data coming in
literally like you want to add a new
branch to your phylogenetic tree
the the computational way that that's
calculated is by
switching in and out in and out in and
out each branch to see
the likelihood of the least likely or
the
the best evidence for that tree that's
existence but it literally
makes every single tree and then
calculates that which is computationally
a really intense um and time-consuming
effort so that's why variational base is
so neat and swift in comparison
yep exactly it's these red and blue on
the bottom it's like a continuum of
approaches
uh before enter variational the
continuum being
we have some simple cases where we have
a closed form solution for exactly how
to update
the model's evidence as new data comes
in and then there's sort of this wild
west
where you need to use customized schemes
and sometimes you have to resort to just
extremely costly like almost brute force
like swapping or doing the markov chain
monte carlo
however maybe there's another way
which is what brings us to variational

base

and variational base as the authors

write on the bottom

the trick of variational base and we're
going to come back to the kl divergence

but

here's the trick up front is consists in
letting go

of trying to minimize the kl divergence
in equation 3 directly

shifting the objective to optimizing a
different functional which bounds the
model evidence

so what would you say the variational
trick is

i mean that's it so if you bound the
model evidence

then you like so you know that the
answer can't be outside of these
parameters if you that's like bounding
the model evidence

then you're not calculating or trying to
you know calculate things that are
outside of those

parameters i would say yep and maybe
there's another way to read this
equation on three but

the underlined red part d_{KL}
is the kl divergence so it's about the
divergence between

two different distributions and the
double line

is like what the two distributions are
so that

$p(x|y)$ given y we saw that back here
that's what we want to calculate so even
though we're kind of accelerated on base
we're still back on equation one doing

inference on the world but it's going to look more complex now so we're looking for the difference between that thing that we really want to find out $p(x)$ given y and then the magical q distribution that we're in control of and then we want the divergence between those two distributions to be zero and distance is always positive there's no negative distance and so we just have to minimize this divergence so that we can get at what we really want and again one part we don't know and it's hard and the other part we're totally in control of so the whole question and trick is going to be can we design a queue so that by reducing the KL divergence on q versus p we're getting out what we truly want so that is kind of what threads the needle between these two approaches the the red being like the elegant and simple and effective and then the blue being like the extremely messy and simulation based variational base is elegant and also it helps us potentially deal with complex underlying distributions cool good okay now here we return in five there's d_{KL} between q of what we want to know and p of x given

y

so it's the same as it was on the

previous equation three

and that is going to be

given another equivalency and it's

restated as

uh the natural log of p of y

minus big l of x so

what what do you think about l of x blue

or what

did you see here

um or sorry no worries

okay i'm like i'm lost in the map

i was just like thinking oh you know

there's two lines that's an equal sign

here's one line is a minus and here's

three lines

yeah i was i was lost at the three lines

like what why is that an equal

sign like this yeah i'm confused at that

three triple line

thing um i think three lines can mean

that it

is defined as like it's strictly defined

as or but

let's find out i don't want to say

anything that's incorrect that's what i

was thinking of that

um i'm like no no i'm i'm opting out in

the past yeah

but um the uh

restatement here is this value that we

want to go

low low low with the divergence between

what we can control and what we can't

control we're going to restate it

in terms of two different pieces a sort

of

log uh this is just the likelihood model
of the observations
so this is just y it's p here $p(x)$ given
 y was like
the true x given the y observation
like the true temperature given the
thermometer and here we're just going to
have
the likelihood of the
measurement so this is a lot easier to
calculate than x given y
because it's not a conditional and then
we're going to
penalize that by $\frac{1}{2} \sigma^2$ of x
which is an expectation of
 q what q thinks is going to be
a likely value for x
plus a joint distribution of x and y
of p not the conditional so
it's a little past me y reframing it in
each of these nuanced ways
helps you do it but i just think of it
like mathematical pinball
like you gotta like bring some wall down
so that you can go up another ramp
and do something different but let's
definitely ask the authors about it
and this the negative is like baffling
like i think like the point of
when you are getting into the logarithm
the logarithmic scale like
so that you like eliminate any
possibility of a negative and then like
they put the negative on the front of it
i'm like
that's a little strange to me yep and
then also it's like
the it's like the negative variational

free energy
and so
we push on though um
they are able to uh
say that basically free energy this is
by analogy to the variational free
energy
they're going to bring that's where they
bring in the f they say basically the
free energy
that you might want to minimize we'll
get there in a few equations
but the free energy function of x that
hidden state of the world like which
model is best or what's the real
temperature
it is the negative of this I
because here what I was defined this way
and it's almost like we we don't we can
separate out
 I from p at all so here
we had like sort of a hybrid with q 's
and p 's
and here there's like a p here but now
when we
get I all alone now we've kind of
encapsulated p in a
way where it's like all q centric i'm
not even sure if that's true so i don't
go too down the rabbit hole but now it's
framed in a way
where we can do our work on
cue but um
but but but get to where we want to get
on p
and i think that is the rub that's
what the question is about is okay it's
a approach

and i would love to ask somebody with more variational inference especially actually on this

uh i think these are the ones where we were like blue and i were just like okay these are we're going to raise more questions than we uh answer

but what does it look like to make the variational version of a model and what do you sit down and do

so that you can calculate this or is it all carried out beneath the hood

so uh

minimizing free energy as defined in this way

on the q which is the function that we control

the goal is that that will also get us to the right value on p

the function we don't control so i have my internal model

of temperature and i want to be doing inference on that finding what's the most likely

the most free energy minimized internal model of temperature

and then if that is attuned properly that should be

accurate in terms of the external as well

so in my like limited understanding of making a variational model um i think that

these models instead of like using values

they use distributions right so so

in that's the kind of the the key

difference i think between
um the variational like model
construction versus the
the like non-variational variety of a
bayesian model
in my limited understanding nice point
that previously we were like looking at
passing
uh like a number through a function and
then
when we're going variational we're
talking about like passing functions
through each other
um this one we're not going to go into
the technical details
but it's about mean field approximations
and this is a super interesting topic
because a lot of statistical models will
assume
that there's like a homogeneous field or
that a gas is well mixed
or that all types of a certain
interaction are equally likely to happen
amongst all
molecules bouncing around but because
we're interested in applying
free energy principle not just to the
burning candle or to a chemical reaction
but to
active inference agents to cybernetic
and
goal-oriented agents how does the mean
field approximation
work like if we want to explain culture
is there a mean field approximation for
culture or
how are we going to apply this to the
systems where we know we're kind of

extending
into analogy land
but at the same time maybe reality is
really multi-scale so it's not just
uh an analogy to say like when chris
field talked about quantum
and communication at our scale not just
as bubbling up from electrons
and not just as a metaphor but actually
as like a third way that it really
was based upon those principles at our
scale
that's what this makes me think about
like do we need
are or will there be mean field
approximations at a lower level
and then we're just experiencing a sort
of non-mean field
at our level
and okay 21 anything else say they're
blue
so i think like that the the mean field
approximation
um i i think it's like i don't know
it's it limits
the interactions between different
variables so you're interacting with the
density instead of
each individual variable interacting
with each variable so it's like um
it's like a course it's a course grading
like
having this mean field approximation i
think yep
and we'll return to it when we look at
the um figures related to fristen's
paper
but yes it's related to partitioning

variables and yes
now this part i thought was really
interesting
and kind of like basic statistics so
maybe you'll be familiar
to some but they're seeing it in a new
way
that the difference between the
different relationships that variables
can have
and we're seeing it here written out
with math but we're actually going to
come back to it
with figures and so it will potentially
be
a little bit clearer but there's a few
different situations
the nicest situation for a statistical
inference is when you can cleanly
separate
inference on the two variables so here
it's like p
 x_1 and x_2 the joint distribution of x_1
and x_2
it can be separated very cleanly because
they're
independent so it's like you flip a coin
and you roll a die
the way that you get the probability of
a head and a 4
is basically you just multiply that
probability of head times probability of
getting a four
and so that's the clean case and sort of
almost by analogy that's how i
understand
variational inference like it just clean
the variables cleanly separate

so that you can do restricted optimization on subsets of variables in a little bit more principled way but um back to the stats there's often a time where the joint distribution that you care about is conditionally dependent so conditional dependence conditional independence two sides the same coin it's conditionally independent on or conditionally dependent on a third variable x_3 and so here we're gonna separate x_1 and x_2 just like we did above but now we have like x_1 conditioned on three and x_2 conditioned on three and there's actually all kinds of situations where we're really interested in these interactor variables and so there's there's one reason it might be latent cause estimation like i want to know why is it that some people take this drug and have this side effect and other people don't sometimes you want to know all three factors like how three different nodes of a computer network or genetic network are related so you actually might be interested in how all three of these nodes are linked and then other times the mediator might be useful for like a statistical analysis or like an interaction effect in an anova like okay there's an effect of age and there's an effect of height

but is there a height by age interaction
so those kinds of questions they're not
the that anova questions not in the
bayesian framework that's from the
frequencies framework but it's an
example
of when you want to know about
interactor
variables but it turns out that this is
going to facilitate
the variational inference because you
can like use this hidden variable z
to shield or separate
 x_1 from x_2 and x_2 from x_1
so that's the the move of pearl and then
that's what we're gonna dive into from
then on so we
we went from basically bazon states
bayes on models
to variational bays variational bays
gives us this free energy minimizing way
to work
the variational inference contingent on
a few assumptions that we would love
clarification and education on
and that's going to help us separate out
in a way that's tractable different
parts of the world model that we care
about
so an easy way to think about this
conditional dependence
is is if you're like thinking about the
probability
of mary and mo both eating at the
frontier restaurant in albuquerque
it's independent of one another right
like mary and mo they eat periodically
at the frontier restaurant this is like

an independent probability that they go there whatever and then they have another friend that they go with so mary and mo go with sally they if sally eats at the frontier restaurant then the probability that mary and mo will be there together because they'll all go together is you know they'll all be there at the same time if sally's there because then they'll all go eat together but they're not independently friends they're only friends they only go as a trio when sally's involved nice our variables just hanging out with each other so let's take it from doing inference on like one variable states and the variational and everything to two variables that's with this joint inference and then here we brought in the third variable let's generalize that beyond three because we're not just interested in x_1 and x_2 in their relationship or how three influences it we want to be actually open to all graphical relationships and that just means everything you can draw as a network with nodes and edges is graphical or that's a network analysis so they bring in now the notion of a bayesian network beta and it is defined as containing g and p

so g is sub defined with vertices uh
and edges v and d so g is really like
edges and nodes so g is the network
itself and that's kind of the matrix
form the two space
form that we're familiar with
and then p is as they write a collection
of tables containing
dependencies between the variables as a
set

of matrices with structure but it's kind
of like

g is the social network like who are the
people and what are the edges
and then p is describing their
relationship

in a way and uh

they also then they write the the tables
 p

uh the graph g is often represented by
an adjacency matrix

so that's a you know nodes by nodes and
then the cells are whether they're
connected or not

and that's exactly what we were getting
at with the two spaces so yeah it is
pretty interesting

and then the tables p contain the
factorization

for the joint probability distribution
over the

variables characterized for the uh
directed acyclic graph the dag

and so there's an equation that we're
going to use to

jointly consider like the network nodes
and edges

g and all the dependencies on the

network

p and so you can imagine if there was
like if you had a matrix of 100 people
and these 50 all knew each other like
they had dependencies between each other
and then these 50 people all knew each
other they had dependencies between each
other

we would want a method that could
decompose that

matrix into a more factorizable
potentially nicer to deal with form

okay so here we visually see it

and i think that the circles are all
meant to be complete

but they look like like i was wondering
if it was a dashed line or something but

i think it's all yeah so here we have a
bunch of variables

t w y x u and z

and then the authors very helpfully
go through a few of the kinds of
relationships

that can link motifs of three
variables and so here's w

and y and so it's like notice how w and
y

are unconnected but they both have a
dependency towards x

so it's like w and y are marginally
independent

but only conditionally dependent if x is
observed so if you didn't see

x you wouldn't know that w and y had a
dependency

but then when you do observe x you can
see that oh

x is actually being influenced by both

of them
that's called um the head to head w
and z marginally dependent
because you could tell that potentially
through latent interactor variable x
if you had w and z measured you could
tell that there was an influence
um but you wouldn't know as much about
the influence of w
on z unless you knew about x and then
there's x
and u which is kind of like the
backwards example of w
and y because x and u would be
correlated in the real world
like you'd be pulling out x's and
pulling out u's and they'd be correlated
so they're marginally dependent but then
if you measured y
if you observed y you would make them
conditionally independent
because you'd be able to correct for y
and then
as you pulled out corrected x's and u's
they would be
conditionally independent so the
bayesian graphical form
allows us to go from that p of t w
et cetera et cetera so think about that
there's six variables
that could be six times six variables to
consider like 36
connections but we can factor it
into like the function of z
given x because that's the only thing
that
z is conditioned on x
is conditioned on w and y use uh

u is conditioned on y and then t is an alone variable

so we've gotten all six variables

into a way that is like a lot

sparser it only specifies the parts that

we actually need to care about

in terms of the dependencies that

actually exist so that's kind of the

bayes network

it's just using network thinking to talk

about

the really nuanced relationships between

sets of variables so this could be like

six demographic factors of a person and

z could be the health outcome

and so then we can use this analysis to

do statistical inference

just like we would with any other

technique anything to add on it

blue uh just the final

formula down at the bottom really

specifies uh like it separates out like

b z given x right and so and t is

independent of everything so you can

really see like the

the conditional dependencies down here

in this final equation

x given w and y and then u given y

so yeah it's almost like there's the

graph representation

then there's the adjacency matrix

representation and then here's

an equation representation so it's like

a rosetta

stone if you could have a technique that

works on matrices and then you wanted to

apply it to a network

well then you need to connect networks

and matrices but then you could definitely apply it there

what about the pearl blanket with this cute picture yeah

so the pearl blanket is really um traditionally is described by the set of nodes that shields a variable so in this case variable a that's in the middle from the other nodes in the network so a can't interact with these nodes that are outside of the pearl blanket which is the represented by this circular blanket so a can't interact with these other nodes without going through the nodes that are contained in the circle

yep and if you notice it has uh like the ones that are to the sides of a it's because all these kinds of relationships are relevant so even the one that a it does not directly connected to a but it still is included within the blanket that's kind of important to note all right so now we're gonna get to this figure and it's kind of clever how i think the authors did it because they introduced this in figure two uncolored and then we're gonna talk through some fristen simulations and then active inference in figure six and then we're gonna return to this in figure seven with a new color scheme that will help really make their point clear so here they're taking this simple

example and they're just
giving another example and um
here is their their legend
the grays are the observables so y is
observable
 y_1 y_2 we're observing two things about
the world
measuring two things all the x 's are
hidden states
the arrows are that bayes net directed
acyclic graph
representation the dashed node
is this special node a_{10} that's like our
focal
node and then the thick nodes
are the blankets around x_{10}
so here just like you had a and it was
like its neighborhood
was the blanket here we're gonna have
the pearl blanket
as they write thick lines are used for
nodes constituting the pearl blanket for
selected node x_{10}
depicted here with a dash border so it's
kind of like a legal argument
they're like saying you know uh would
you agree
that this is the appropriate use of the
pearl blanket because if you do think
that this is a pearl blanket
you might find yourself in a trap in
figure seven when they
show you what this actually represents
so it's like
okay good this is what a pearl blanket
is there's no
real dissension this is where we're at
with pearl blanket

thinking and we've just specified some arbitrary network
you know nothing in mind just specified it
and then they also um define in 4.1 uh a little bit about how pearl blankets play a role in active inference and it's a super fascinating topic that is something we want to return to with the authors but if you notice that this q distribution that was the one that we were in control of
and then that minimization previously we were looking at f of x it was free energy minimization on states hypotheses but here we're also including policies π sequences of action so now we're doing free energy minimization on policy and so that's what allows us to include the action in the loop with active inference
so active inference isn't like just a two-stroke engine where we're going to do inference on states and then action selection in a second step it's actually a loop that integrates action and inference into functions in a co-equal way and that's what one of the special pieces of active inference is as they write um something that classical formulations of variational inference in statistics and machine learning do not consider

instead assuming fixed observations or data so you read in the big data set you do the free energy minimization and there's your answer and then if you get new data what do these companies do they re-run the optimization with the new data but this is actually like real time in the loop and it's also taking action okay any thoughts or on figure three do you want to intro it or what yeah so this uh figure i think the papers on the next slide actually the the figure that came out of the um it came out of this first in 2013 paper and there's like a set of figures in this paper that were accepted from that paper and so this is the it's a primordial soup simulation that was done in in the article by firstin and the larger cyan dots represent the location of each particle and then there are three smaller blue dots associated with each particle representing the electrochemical state of the particle so this is like on on how life came to came to kind of be right so that's the the what's going to happen in the simulation so this is just the first and then we go on into the into the next

figures from the paper

nice so take this one yeah

yep yep so the simulation is just like
the cyan it's kind of like the nucleus
of an atom

and then the blue dots the little ones
are like electrons floating around

okay so just it's it's not trying to be
a physics simulation but that's kind of
what's happening

figure four um

here is oh and and they're electrically
charged

that's the electrochemical part here's
what they write

um you can take so let's just say each
of these cyan dots had a number
you know number one through 100 like an
agent-based model

now we're going to element 1 through 120
i guess

you can plot the adjacency matrix a
based upon the coupling that is
dependencies

between the different particles at the
final simulation t

so you're going to start this simulation
let it run out

and then okay this one is very closely
coupled to that one and this one is
closely coupled to that one so

number one is talking to four and number
four's talking to 22.

and then you can take that matrix and
you can just put a dot where you saw a
dependency

so if you saw some sort of behavioral
dependency between the two like in

adjacency

you're gonna add a dot and then you're
gonna rearrange

the rows and the columns by swapping
which preserves the information

rearrange them

so that they're sort of uh falling out

there's actually several there's there's

a pattern that falls out

we see that there's a broad region of
interactions over here

and then in the bottom right there's
like another

kind of sub area of interaction and then

on the very bottom right is like a very
tight

area of interaction where you see a
bunch of dots over the 120.

so as they write the adjacency matrix is
a representation

of the electrochemical interactions

between particles because it's drawn

because the adjacency is on the

simulation but can be interpreted as an

abstract depiction of a bayesian network

and so we're taking the dependencies

between these particles as they

jostle around and this is what we're

trying to do we're trying to almost like

factorize it separate nature at the

joints

you know map in the territory okay

there's a border on the map

but is there a border on the territory

and then

um plot it as a matrix and then

there's some shading that's going to be
suggestive

of what role those
nodes are hypothesized to be playing so
here's what they did just in a little
more formal
on slide 30. they used spectral well
this is
i think what first in 2013 did yeah uh
spectral graph theory was used
to identify the eight most densely
coupled nodes which are defined as the
internal states
so y eight y dense cup okay
right that's the question that's what
we're discussing
and those are going to be defined as
internal states
just to skip to the slide those are
those blue ones right in the middle
here so those are those core states back
here given those internal states
the markov blanket is then found through
tracing the parents
children and co-parents of children in
the network
sounds like a preschool
um as an extra interpretive step and
that's
part of what this discussion is about
how interpretive is this like
interpretive dance interpretive
or like interpreter at the un um
the nodes in this mark called blanket
can be further separated
into sensory and active states the
sensory states correspond to the parents
of the internal states
so that's incoming info not that they
birthed the internal states but the

sense states are the ones that have an upstream dependency on the internal states

and the active states correspond to the children of the internal states and their co-parents

so again not that the active states are simply birthed by internal states but that they have downward flowing statistical dependency

states that are not internal states or part of the markov blanket are then called external states

so we're using this partitioning where first you identify dense couplings then traced out parents and co-parents and children

and then everything that's left is external states and so you recolor that simulation from figure three

here it is uncolored and now we're gonna recolor it

with this new color code where we've identified

the blue densely coupled nodes and then the active states

and then the um sensory states which are sort of on a frontier

with the uh teal states the cyan the external so it's almost like you have the internal you know let's just imagine it's like

the nervous system and then it's swaddled in the active state

and then on the outside of the sense of the active states or the sensory states and those interface directly with the

world
so that's kind of the tantalizing
nature of this model would be like wow
if it really just fell out
of a simulation that you can get
self-organization like that
wouldn't that be cool but what is the
issue here blue
well i mean it's kind of arbitrary um
like so you know they said a third of
the
of the electrochemical states from the
very beginning of the simulation
were not did not interact right so it's
that establishes this non-equilibrium
system um and and i think that
uh you know the the issue then comes
from
assigning these interactions to
internal external active and sensory
states that may or may not
have some kind of dependence on
one another yes
as they write here the simulation
assumes
that by viewing the system through the
formalism of the markov blanket
plus some additional assumptions about
the separation of states the ones that
we kind of walked through just a second
ago
you can uncover hidden properties of the
target system
which is then said to quote instantiate
or possess a markov blanket
and then this is the map territory the
procedure of attributing to the
territory which is the actual dynamical

system

what is a property of the map which is
the scientist doing inference on the
dependencies and then

cutting it at eight nodes and then using
this other approach

to make the markov blanket it's a
property of the map

and it's a clear example of the
reification fallacy

treating something abstract as something
concrete without further justification

that's not the only way reification
works but it's one key piece of it and

so in other words it's like you go in
with a marcotte blanket hammer

and your secondary pipeline for
identifying parents and children

and then you find it because it's
defined for any system you're going to
be able to get a markov blanket using
that protocol

and then look at this have you really
discovered something about this
dynamical system by separating it into
this partition when actually all of the
particles

are doing the same jostling motion so
it's not like we have uh the influencer
nodes

here these are just densely coupled ones
and so

um they propose

in this same paragraph proposed to
distinguish between pearl blankets
to refer to the standard epistemic or
like statistical

use of the markov blankets and first in

blankets to refer to this new
metaphysical construct
and they also give a shout out to martin
beale in the paper
and so we're kind of coming back to this
map territory distinction
on the map side we have epistemic
pearl blankets bayesian statistics
but no philosophical power and then
what they're attributing to the
territory
would be metaphysics and it would be the
kinds of things that the fristen blanket
is making claims about
and i think my question was
can we do that action in the loop
without
metaphysics because it seems like
there's two
innovations in or differences between
the
pearl blanket in its sort of normal
statistical use and the way frison's
using it
first he separates sense and action but
also he's making metaphysical or
ontological claims
so i'm just wondering like are the
ontological claims
separable from taking a control theory
perspective on bayesian statistics
that's what we would like to find out
so then blue you copied this image over
what do you like about
it or what did you see in it so i just
thought it was like a nice
pictorial representation as opposed to
this primordial soup that gets a little

bit hairball
of what we have as internal states
which were back in the in the christian
picture the the central
um dark blue most highly connected nodes
so we have the internal states and then
the external states
which are separated from these blanket
states
and so that's this where this pink
dashed line is surrounded
that that's labeled with the bees so
this is like the fristen blanket
contains the active states and the
sensory states
that may be interacting with each other
uh and then this fristen blanket is what
separates
the internal states from the external
states
um and they take it in quite a literal
sense like the boundary of a cell
and and um so forth so separating the
the internal from the external is the
first and blanket
nice and let's connect this to figure
four and then the kinds of things that
we wanted to discover there
looking at this as a graph of bayesian
variables
it's like active states they influence
external states
so this is that coupling so it's almost
like external states
they don't connect to internal states at
all that's kind of like
missing a corner on the graph so we have
nodes everywhere you know there's cells

in your hand and there's cells across a wall
but how can we separate nature at the joints
maybe we could make measurements
find the statistical dependencies and then
carve those dependencies into densely connected internal states
and then incoming states that are upstream and outgoing states that are downstream and then we might have a new way
to do bayesian modeling on systems that are on a continuum
or as part of a really uh complex process
and figure five yep we already had that
i guess we put it
put it twice
and yes 4.3
this is sort of the uh
man's search for meaning why and how
have markov blankets
been reified to act as parts of the target system
how did we get here and uh
where this technical idea became reified
so people thinking that there's support in the literature
saying oh yeah markov blankets it's in the literature there's real measurement support for markov blankets
that's reification um of delineating spatial temporal boundaries
rather than formal tools intended for scientific representation and statistical analysis

when did the map become conflated with
the territory
because like everyone's familiar how we
use linear regression
on healthcare data but then no one goes
back and says well then you're
a linear regression so it's a little bit
like
if somebody were using markov blankets
on health data
and then said now you're a markov
blanket
or is it i'm not saying it's exactly the
same but that's the rhetoric
of what is being uh put out here
it would be like the line delineating
the healthy from the sick
is a markov blanket it's a line
there's actually a line
and then they write
so again people should read these
papers they're fun to think about and
there's a lot more that we're not
getting to
go into in the um zero but they write
basically
perhaps surprisingly many authors in the
field are seemingly not aware of this
process of reification
and this has led to the conflation of
several different kinds of boundaries in
the literature
markov blankets are characterized
alternatively as
statistical boundaries causal boundaries
spatial boundaries epistemic boundaries
and autopoietic
boundaries each characterization is

treated as somehow equivalent to
and interchangeable with the others now
it's rare that someone does the grand
slam
where they totally conflates all of them
but then
there are papers here and there where oh
it's causal and it's spatial
and then it's spatial and it's epistemic
and it's statistical and it's causal
and through this web of formalisms
we can get in a tangle
to kind of any thoughts about that yeah
just that the paper itself contains like
many quotes
from other papers where they use the
markov blanket or the fristen blanket
in very much um like a statistical
and um instrumentalist sense and then
turn around and use it in a very
realistic sense
so there's many uh like examples of that
illustrated through the paper that trace
back through the literature good
point and those are the parts that we
didn't quote
in this dot zero but there's many pages
so if someone's reading it and they're
like
wait they don't do that in the
literature read the paper
because that's where they mention it and
this is from
the live stream number 15 where we had
uh we were discussing the paper of fan s
and hippolyto
and we introduced this discussion around
realism and instrumentalism

and what those authors had written was
basically realism and instrumentalism
concern the models and statistical
manipulations that make up the fep
whether they are thought to be used and
manipulated by the systems under
scrutiny

independent of scientific inquiry

realism

the tree falls in the forest does it
make a sound

realism yes there's a real tree it
really did fall it really did make a
sound

or conversely whether they are thought
to be scientific tools wrought by humans
in specific socio-cultural environments
to study particular systems

instrumentalism

so that's like well no a sound means
somebody has to hear it so
you don't know if the tree is real or it
really didn't make a sound

and the whole question is where does
free energy principle

is it about what systems are doing
realism when we don't look

or is it about us modeling systems
no matter how they look that's
instrumentalism

and that brings us to figure six
where they show a classic sensory motor
loop

so it looks like it's an active
inference loop and that's where they're
going with this

but their whole point is that um
it although the figure does not use

directed edges to signify causal influence

it is not strictly a bayesian graph as it depicts cyclic sets of circular dependencies

some between pairs and an overall loop compo containing all components

now this was quite interesting because first off

it points out that yeah octave inference didn't invent

the internal external or the action sense idea

we're using different models and we're recombining them

but there is this cycle right there's also

in active inference more of an emphasis on the internal states and the external states

as they have their own endogenous dynamics but that could be part of these models too

and then it's true that it's not a directed acyclic graph if there are loops

local or global loops but what about implementations

that have time steps where you can break these loops like if you compute it in this order

just a thought like does it really violate how how badly does it violate some of the assumptions of directed acyclic graphs

but i don't know any thoughts are good for seven

i don't know yeah i don't have the

answer for you if i did i mean i don't
know that necessarily putting a blanket
over the sensation and the action
states the sensory in the active states
like putting that in a box like does
that

break the acyclic graph thing i mean i
don't know that that necessarily
breaks it enough to to make it somehow
valid either

right like i don't know it is a cycle

i think good question so now

we return to the figure that we saw
before

and this is where they now strike so
figure two remember it was like surely
you agree this is how a pearl blanket
would be deployed on this

graph now speak now or forever hold your
peace

and then now it's going to come back to
bite some people who may have not
expected where it was going to come
so before they were highlighting that
x10

node and now they're going to color
according to

fristen's scheme they're going to color
the notes

so it's colored like figure 5

but on the figure that they introduced
earlier

with x10 still as a focal node

but then they're going to show that if
you have a different focal node

x9 not only are different nodes included
but they're just they're categorized
differently

which isn't a problem for the pearl
blanket
because it's like oh if you if you're uh
if you measure
variable a and b then this one is the
child or the parent depending on you
know what you measured in that bigger
graph and then but if you're going to
have a realist interpretation
of what these nodes are you're out of
luck
because one is going to be an active
state for one in a sensory state for
another
and then if you are going to say well it
doesn't matter if it's an active state
or a sensory state like they're all
doing it for each
other it's equivalent to the pearl pearl
blanket
phrasing so what they write is if you're
going to have a sensory motor
interpretation
which is not the only option this
suggests that the sensory motor blankets
are markov blankets applied under
specific set of assumptions
and it can't be traced to standard uses
of markov blankets in variational
inference so basically depending on
what your hidden variable of interest is
you compute
a different blanket around it so that
is problematic for realist
interpretations because depending on
what you're doing inference on
you might variously find that the
membrane of a cell

or a measurement from the membrane of a
cell was a sensory state
because it was incoming for something
else but it could also be an
action state because it's outgoing of
something else but then it would be
internal
to another set of things and so if we
want to make realist claims or
interpretations
about blanket states and internal
external states
that's an issue
so is the issue like that we haven't
figured figured out how to like nest
these models
yet i mean i i think that it makes sense
that the outgoing is
incoming for another agent
right in the in the model so that we
just haven't figured out the nesting
pro properly yet appropriately or like
that each nest is its own
like blanket state i don't know
i agree i'm an optimist as well
let's see eight but this is this is
where they
they really go for it and this one i
thought was just very clever and funny
so
the delayed figure you know with the
intermission to convince you
lull you now eight so now they're going
to go into
a graph that oh total coincidence it
looks like the left side
so here on the right side you know we
were talking oh x10

x9 but we never talked about this left
one that looks like
california but let's look at it now
[Laughter]
so california exactly um on the left
a bayesian network where i_d and $i_{sub\ p}$
denote the motor intentions of a doctor
and the patient respectively
so that's this these ones on the top too
h
is a medical intervention with a hammer
c
is a cortical motor column and s are
spinal neurons
movement is m and then k is a different
way
that the leg could have been observed to
move like somebody could have moved it
or gravity could have changed so again
m is the movement that's the observable
movement
this is the abstract bayesian net on the
left
in a in b is
one possible action pattern which is
that the
the patient goes to the
cortical motor command the ip
and then h is the medical intervention
with a hammer
and then that results in no
motor behavior but then here
you do have um the motor behavior
observed
but it comes about a different way
rather than h
the intervention as a co-parent you have
k

this other way like somebody picking up
the leg and moving it in the middle
and on the right the same network is
partitioned using a naive
first and blanket with different choice
of internal states c
and s respectively so if your internal
state is the cortical
neurons you're going to
think that the uh basically active
states are the spinal neurons
which is how people talk about it so
it's not like this is like a
the theory is blown out or anything it's
like saying the spinal neurons
are the active states of the cortical
neurons
and then the movement is if your focal
node internal is the spinal neurons
then the active state is what we would
call
in realism the real active state
so it's kind of like saying again
depending on what node you're focused on
different nodes are going to be
categorized as a sense
active internal and external i want to
say that's how we've been thinking about
it
that depending on which side of the line
you're on that's going to be internal
external
but the point is it provides um problems
for realist interpretations of any
single markov blanket
because we have to acknowledge that the
markov blankets and especially how the
states are

classified is conditional so it's like
is the bicep an active state or is this
an active state it's like well if you're
doing inference on this
yes but if you're doing inference on
that no
so it's hard to have a realist
interpretation
even though there are times where if you
define something
internally as something that's truly on
the inside of something
it might be that the active states truly
line up with what's on the outside of it
but that isn't the full generality
of this bayesian network approach and
that's like a broken clock is right
twice a day
and it doesn't actually get as general
as it could be
this makes me think of like the broken
knee like the phantom limb
it takes me back to that like what if
the you know sensory
neurons are activated for the leg to
move but there is no leg to move
right like then what nice
so okay next um one
uh they give a little bit of recent
history
so if twitter's too fast and textbooks
are too slow
this is the the history speed for you
beal pollock and canai 2020 questioned
some of the technical assumptions
underlying the use of markov blankets by
fristen
and since then the idea of the fristen

blanket being understood as a distinct
construct has gained traction in
literature

and we're actually looking forward to
having martin

discuss that with us in about two weeks
so again it will be really interesting
to hear like

how did you come to the position where
you were getting at these technicalities
and critiquing it and

where does action fit in where does
philosophy fit in

where do these other things that fristen
is bringing

in and um

[Music]

any anything to say on this this part or
only the question that i raised at the
beginning where does language begin
right like to what to what extent is the
the problem

or or the issue that's being raised like
a function of the language that we're
using

yes and in this paragraph they make
another

subtle point that shows why when the
language is uncertain there can be a lot
of drift

underlying they say in a recent paper

fristen blankets are formalized

in terms of constraints on sparse

couplings and then they're identified

using read the technical details

the non-zero components of the Hessians

using some other

last names and that is taking it the

construct according to the authors
far away from the pearl blanket origins
and so they're saying you wouldn't make
that kind of swap
in the underlying mechanics unless you
were working
with the physical domain of what's
really there
um so that's an interesting point would
be cool to hear from the authors
here on the next slide is a kind of a
closing
remark by the authors
and this is kind of a meta point and the
meta point is actually
the relationship between action and
inference
like of an organism and then
scientists modeling their world so
it's not the first person who's made
that comparison you know
you got physical foraging you got
information foraging but they're saying
basically
there's a lot of similarities between
perceptual inference and scientific
inference
both use a previously learned model of
the world and set of observations to
infer the causal structure of the
unobserved outside world
and this was like the triple meta is
model-based cognitive neuroscience is in
a special place
it makes models of how animals model
their environments
a cognitive neuroscientist uses both
behavioral and neural data to infer the

most likely model that the agent's brain implements there's nothing wrong with this doubling up of modeling relations as long as one is conceptually careful one needs to distinguish between the properties of the environment properties of the agent's model of the environment and properties of a scientist

model of the agent modeling its environment and that's the third level and that's why it's unsurprising actually almost in retrospect that fep arose from cognitive neuroscience and neuroimaging with eeg and fmri because it doesn't come from we're going to use physics to find the grand unifying theory it actually came from a we're making measurements about how agents reduce their uncertainty about their model of the world and it's that that gives you the triple loop which i think is like quite interesting about how it's nested well and here we are adding like another layer to the loop like thinking about how like trying to you know criticizing the model with which the scientists are using to model the agents it's just models all the way down yep so what are the implications for cognitive science in humans and this part right here properties of how the scientists

model their agent modeling their
environment you could replace
scientists with researcher or
data scientist or machine learning
algorithm
this really matters what people take
into account
when they design their algorithms and
when they make algorithms about other
cognitive agents
because that constrains
in ways that are sometimes obvious and
sometimes not obvious
you know if you only had one variable
for a user on your website how likely
they are to buy something
that was your variable you're going to
set up a different algorithm
so it's the properties of how the
scientists models the situation that
really matter
and um has implications for cognitive
science that when we're studying humans
and non-humans
and non-animals
and blue's big question
this is actually the big question that
the authors bring so and so they asked
in the paper
if bayesian networks and markov blankets
are the right kinds of conceptual tools
to delineate the sensory motor
boundaries of agents and
living organisms so the authors you know
brought this up
within the paper i thought it was just
important to highlight that
because i don't know i think like if

we're
you know if you go out to a scale like
if you
have an image and you start to you know
coarse grain that image you know
building representations of the image
building representations of the
representations of the representations
of the representations eventually you
can't see the picture anymore at all
and the point is to make it easier to
understand not to
you know make it so blurry that nobody
understands it at all i think
cool this is what brings them to
the distinction of inference with
uh wait is that even the right it's uh
yeah oh yeah inference width and
inference within i was thinking
i was getting really you know inference
of a model but
inference with a model the distinction
between the two blanket constructs as
they're positing them
i'll shrink our little video can then be
easily identified
once we look at who appears to be
performing the inference and
what system that inference is performed
on
so they go through four kinds of
networks we're not gonna
go through it here but if people are
interested
in the technicalities here it's really
worth thinking about what's the
difference between the generative model
and the generative process so like i can

have a generative model of blue
in my head so when she's speaking i'm
doing generative inference
on her syntax but then that's not the
generative process which is the actual
underlying person so this is just a nice
very um rigorous way to break down
when we're doing inference on a model
versus inference on a
on a process and
okay 42 all of this points towards
a fundamental dilemma for anyone wanting
to use markov blankets
to make substantial philosophical claims
about biological and cognitive systems
which is what we take proponents of free
energy principle to be wanting to do
so agree or disagree you know if you
disagree you can write a paper or you
can come on
activestream but that's the claim
of the authors and so that is what we
kind of got excited to learn about
and they then come back to this with
versus within
and they say basically in inference with
a model
the graphical model is an epistemic tool
so that's like i'm going out to the
forest with my machete
i'm gonna use it as a tool as i go out
versus inference within a model the
scientist disappears from a scene
becoming a mere spectator of the
unraveling inference show
before their eyes the way that's written
makes um the authors sound a little
bit um sort of dismissive of that being

a likely approach

it's almost like well we wouldn't want

that to happen or it can't happen

so scientists can only do inference

with models now the scientist as the

body

is the inference within a model like the

inference is happening within the

science body or within the scientists

exocortex or within their community

but that's very different when you're

talking about the real

system the scientist at the desk that's

the system that is doing inference

but inference with a model is the

instrumentalist take

okay and um

this is just a final slide that just

gives some closing thoughts

and this was interesting because they

mentioned that

there are there are simulations like the

game of life

which have been enormously productive

in many domains and so basically they

say that it might be an interesting way

of

modeling emergent processes in complex

systems but it won't

support any metaphysical claims about

first and blankets and then they say we

will not pursue this idea any further

here

but offer it as a more modest perhaps

instrumental interpretation

that some proponents of the fep and

active inference might

be inclined to adopt in order to avoid

any stronger commitments
so that's in some ways where we
feel like we're sitting which is we're
approaching active inference
instrumentally
and then we'll see what happens with the
other claims but it seems like we can
pursue it instrumentally
and i guess that's a modest proposal
and then they just summarize so overall
nice paper relatively
good length a lot of figures clear
arguments rigorous philosophy and
math so it was um good times
and like a good coverage of the
literature and um
you know a trace through the history of
the the
methods underlying the fep was good do
you have any other
closing thoughts only that
maybe inference within a model is
actually possible if you're
using a robot that's programmed to do
inference
right like the inference if the if the
or if the
action is to do inference if that's how
it works then then that's inference
within a model
possible cool
well it will definitely be really
interesting to
hear from the authors hear from
everybody who wants to join
on the stream so blue thanks a lot for
joining this was a great discussion
thanks everyone for watching and we'll

see you later

bye bye thank you